gRPC + Protocol Buffers

What is gRPC?

gRPC is a **high-performance, cross-language, open-source framework** for **remote procedure calls (RPC)**. Created by Google, built on:

- Faster and more compact than REST/JSON
- Strongly typed interfaces
- Automatic client/server code generation
- First-class interop (Java, Go, Python, C++, C#, Kotlin...)

When to use

- Microservice-to-microservice communication (low-latency)
- High-throughput services
- IoT / robotics / Edge computing
- Strongly typed enterprise systems

Not ideal for

- Public APIs (JSON is easier)
- Browsers (no direct gRPC support)
- Simple CRUD applications

Protocol Buffers (Protobuf)

A **binary serialization format** with:

- Very compact data representation
- Code generation for strongly typed languages
- Strict backward/forward compatibility rules
- Schema defined in `.proto` files

Why Protobuf is Fast

Compact Binary Encoding

- Field names **not included** → only numeric tags
- Variable-length integer encoding
- Very small payloads (5–10× smaller than JSON)

Efficient Parsing

- Messages mapped to classes (precompiled code)
- Zero-copy deserialization patterns available

---

gRPC Service Definition

```
syntax = "proto3";

package user;

// ----------------------
// Messages (Entities)
// ----------------------

message Post {
  string timestamp = 1;
```

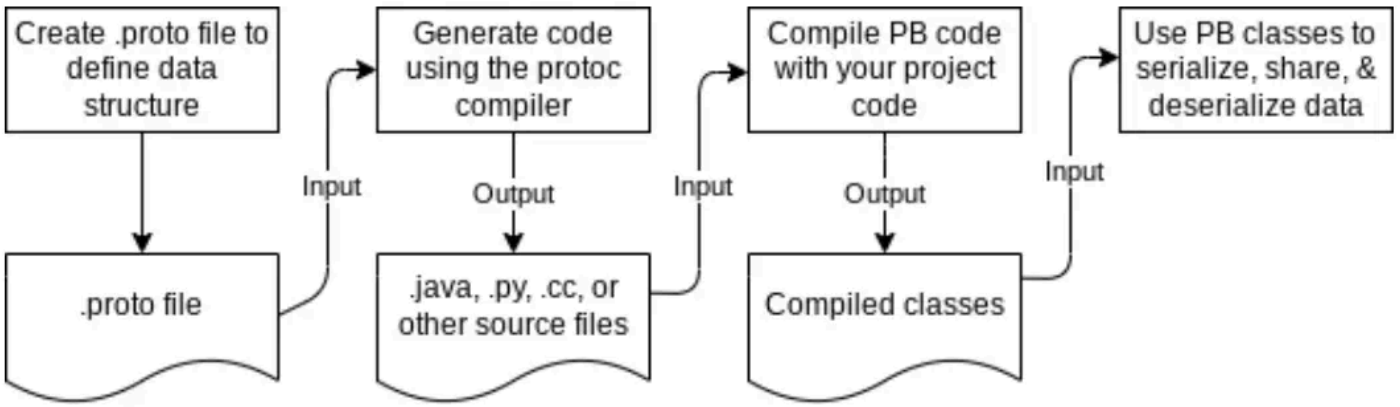
```
  string content   = 2;
}

message User {
  string userUUID  = 1;
  string nickname  = 2;
  string birthDate = 3;
  repeated Post posts = 4; // a user can have multiple posts
}

// Request and Response messages
message UserRequest {
  string userUUID = 1;
}

message UserResponse {
  User user = 1;
}

// ----------------------
// Service Definition
// ----------------------

service UserService {
  // Get a user by UUID
  rpc GetUser (UserRequest) returns (UserResponse);

  // Add a new user
  rpc AddUser (User) returns (UserResponse);
}
```

Client/Server Generation

```
protoc --java_out=. --grpc-java_out=. user.proto
```

Generates:

- Server base classes
- Message classes



Schema Evolution in Protobuf

Compatible Changes

- Adding optional fields

- Renaming a field (if tag number stays the same)
- Removing a field (keep the tag reserved)
- Changing default values (clients unaffected)

Breaking Changes

- Changing a field’s type
- Reusing tag numbers
- Changing field labels (`repeated` → non-repeated)
- Removing fields without reserving tags

---

gRPC in Java (Spring Boot)

Server Implementation

```
@GrpcService
public class UserServiceImpl extends UserServiceGrpc.UserServiceImplBase {
    @Override
    public void getUser(UserRequest req, StreamObserver<UserResponse> res) {
        UserResponse response = ...
        res.onNext(response);
        res.onCompleted();
    }
}
```

Client Usage

```
ManagedChannel channel = ManagedChannelBuilder
    .forAddress("localhost", 50051)
    .usePlaintext()
    .build();

UserServiceGrpc.UserServiceBlockingStub client =
    UserServiceGrpc.newBlockingStub(channel);

UserResponse res = client.getUser(
    UserRequest.newBuilder().setId("123").build()
);
```

Resources